Classical Demand II: Duality and Slutsky

Advanced Microeconomics 1: Economic Decision Making. Lecture 3

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Where we left the question

- **Lecture 2 built the ideal answer**: the correct indexation payment for household h is $e^h(p^1, u^0) - w$.
- **And admitted the catch**: e, h, v are all constructed from u — and no dataset contains u. The answer is ideal and uncomputable.
- **What data do contain**: Marshallian demand $x(p, w)$ — purchases at prices and budgets. Supermarket scanners, the Household Budget Survey, your quiz.

Today: four bridges from x to the unobservables — Shephard, the second-derivative structure, the Slutsky equation, Roy. All four are derivative identities. By the end, the size of the indexation error is a formula in estimable objects.

The one idea behind all four bridges

$e(p, u)$ and $v(p, w)$ are **value functions** — the optimized worth of a problem, as its parameters move.

- Differentiating a value function is easier than it looks. When p_ℓ ticks up, two things happen: the old plan costs more (*direct effect*), and the plan re-optimizes (*indirect effect*).
- At an optimum, the indirect effect is second-order: the plan was already the best available, so small adjustments to it are worth nothing at the margin.
- So only the direct effect survives — and the direct effect is always something *simple*. This is the envelope logic; the next four theorems are four instances of it.

Bridge 1: the index’s slope is a demand

Shephard's lemma

Maintained: u continuous, representing LNS and strictly convex $\succsim$; h single-valued, differentiable where needed.

Theorem · Shephard's lemma (MWG 3.G.1)

For all $p \gg 0$ and u: $$h(p,u) \;=\; \nabla_p\, e(p,u).$$ Proof

- Envelope logic in one line: raise p_ℓ by a cent, and the cost of staying as well off rises by — how much you buy of good ℓ. Substitution is second-order.
- Read it for the thread: the true cost-of-living index's slope is a demand. The unobservable e has an observable derivative. First contact between the ideal answer and the data.

Differentiate once more

Theorem · MWG 3.G.2

With $h(\cdot, u)$ continuously differentiable at (p, u), the $L \times L$ matrix $D_p h(p, u)$ satisfies:

(i) $D_p h(p, u) = D_p^2 e(p, u)$;

(ii) $D_p h(p, u)$ is **negative semidefinite** and **symmetric**;

(iii) $D_p h(p, u)\, p = 0$.

Proof

- (ii) NSD is the differential compensated law of demand — and it is exactly the concavity of e, Lecture 2's sleeper property, now cashed in.
- (ii) **symmetry** is new information: cross-effects of compensated demand are equal, $\partial h_\ell / \partial p_k = \partial h_k / \partial p_\ell$ — Young's theorem applied to e. Hold this thought two slides.

Bridge 2: the meeting point

The Slutsky equation

Theorem · Slutsky equation (MWG 3.G.3)

For all (p, w) and $u = v(p, w)$:

$$\underbrace{\frac{\partial h_\ell(p,u)}{\partial p_k}}_{\text{unobservable}} = \underbrace{\frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w} x_k(p,w)}_{\text{estimable from demand data}}$$

Proof

- The right side is *exactly* Lecture 1's S — built from choices, no utility anywhere. The two constructions of the course meet: $D_p h(p, u) = S(p, w)$.
- Extremely important empirically: h is unobservable, yet its derivative matrix is computable from Marshallian data — every applied demand system lives here (Deaton & Muellbauer 1980).

What the meeting point settles

- **Preference maximization** $\Rightarrow$ S **symmetric, NSD,** $Sp = 0$. Lecture 1 got NSD $+$ singularity from WARP alone and flagged symmetry as *conspicuously missing*. Now we see what fills the gap: symmetry is the demand-level fingerprint of transitivity — one coherent ranking behind all choices.
- So the preference-based approach imposes *strictly more* structure on data than WARP: the gap between the two routes is exactly one matrix condition. Testable — econometricians test it (and Lecture 4 shows why a rejection would kill the compensation program).
- **Decomposition.** Rearranged: $\dfrac{\partial x_\ell}{\partial p_k} = \dfrac{\partial h_\ell}{\partial p_k} - \dfrac{\partial x_\ell}{\partial w} x_k$ — every price response splits into *substitution* (signed, by the compensated law) and *income* (signed by normality). Giffen goods decoded: $\partial x_\ell / \partial p_\ell > 0$ requires an inferior good whose income effect overwhelms substitution.

Substitutes and complements, finally well-posed

Definition · net substitutes / complements

Goods ℓ and k are **(net) substitutes** if $\partial h_\ell / \partial p_k \geq 0$ and **(net) complements** if < 0.

- Defined on *Hicksian* demand deliberately: symmetry makes the relation mutual (ℓ substitutes for $k \iff k$ for ℓ), which the Marshallian "gross" notion fails — asymmetric income effects can make ℓ a gross substitute for k but not conversely.
- From $D_p h\, p = 0$ and $\partial h_\ell / \partial p_\ell \leq 0$: every good has at least one net substitute. Substitution is not optional in this theory — which is precisely why fixed baskets cannot be exactly right.
- The IOU from Lecture 1's comparative statics section: paid.

Bridge 3: from v back to x — and back again

Roy's identity

Theorem · Roy's identity (MWG 3.G.4)

If v is differentiable at $(p, w) \gg 0$, then for every ℓ:

$$x_\ell(p, w) \;=\; -\frac{\partial v(p, w)/\partial p_\ell}{\partial v(p, w)/\partial w}.$$

Proof

- Forward reading: demand from v by differentiation — far easier than solving first-order conditions. The workhorse demand systems (AIDS, QUAIDS) are *born* this way: posit a flexible v, differentiate.
- Backward reading (ours): Roy is a differential equation linking an *estimated* $x(p, w)$ to the unknown v. Solve it, and the data hand you indirect utility — hence e, hence the ideal answer. This is precisely Hausman's (1981) engine, executed in Lecture 4.

The answer, in estimable form

CV and EV are areas under Hicksian demand

One price changes: $p_1^0 \to p_1^1$, others fixed at p_{-1}. Since $h_1 = \partial e / \partial p_1$ (Shephard), the fundamental theorem of calculus gives

$$-CV = e(p^1, u^0) - e(p^0, u^0) = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^0)\, dt, \qquad -EV = \int_{p_1^0}^{p_1^1} h_1(t, p_{-1}, u^1)\, dt.$$

- The ideal answer is **an area under a compensated demand curve** — u^0-curve for CV, u^1-curve for EV. They bracket the Marshallian curve for a normal good.
- And the integrand is reachable: the Slutsky equation delivers h's derivatives from estimable objects, Roy delivers v from an estimated x. The uncomputable answer of Lecture 2 is now a computation.

The over-payment, measured

Second-order Taylor expansion of $e(\cdot, u^0)$ around p^0, using Shephard ($\nabla_p e = h = x^0$ at (p^0, u^0)) and $D_p^2 e = S$:

$$e(p^1, u^0) \;\approx\; \underbrace{p^1 \cdot x^0}_{\text{basket payment}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta p^{\mathsf{T}} S(p^0, w)\,\Delta p}_{\leq 0 \text{ (NSD)}}$$

- So the **substitution bias** of basket indexation is $\Delta p^{\mathsf{T}} S \Delta p/2$ in euros — Lecture 1's inequality, now a *formula*, in the matrix Lecture 1 built from choices alone.
- Check against Lecture 2's toy: poor Cobb–Douglas household, energy $+50\%$ — exact bias 0.9pp of budget; this quadratic approximates it to first digits. The bigger and the more *relative* the price change, the bigger the bias: 2022 was maximally both.
- Measured in the wild: over the UK's 2021–23 surge, substitution cut the cost of maintaining pre-surge living standards by 1.5pp against a 26.2% Laspeyres rise (Chen, Levell & O'Connell 2025). Lecture 4 uses their full accounting.

When the simple area is legal

Practitioners often report *Marshallian* consumer surplus: $\Delta CS = \int_{p_1^1}^{p_1^0} x_1(t, p_{-1}, w)\, dt$ — no u anywhere, one estimated curve. When is that the ideal answer rather than a third, wrong number?

Theorem · no wealth effects (Marshall's case)

If $\partial x_1(p, w)/\partial w = 0$ for all (p, w) in the relevant range, then

$$CV = \Delta CS = EV.$$

Proof

- Plausible when the good is a small budget share — and *that is an empirical statement about who you are*: energy is small for the rich household, a tenth of the budget for the poor one. The shortcut's legality varies across the income distribution — one more reason one number for everyone misleads.

Where the question stands

Takeaways

- **Four bridges built**: $h = \nabla_p e$ (Shephard); $D_p h = D_p^2 e$ symmetric NSD; $D_p h = S$ (Slutsky) — the course's two constructions are one; x from v (Roy), and v from x read backwards.
- **The answer is estimable**: CV/EV are areas under Hicksian curves; the substitution bias of basket indexation is $\Delta p^{\mathsf{T}} S \Delta p / 2$; the CS shortcut is licensed exactly when wealth effects vanish.
- **One debt outstanding**: all of this assumed a rational household *exists* behind the data. When does observed demand actually pin one down — and what certifies your 25 quiz choices? Lecture 4: integrability, SARP, Afriat.

Required: MWG 3.G; skim 3.H before next time.
Papers: finish Hausman (AER 1981) — Lecture 4 walks its pipeline on the thread.
Ahead: Lecture 4 on Thursday; the practical on Monday — laptop, working R, your quiz ID.

Appendix: proofs

Proof: Shephard's lemma (MWG 3.G.1)

Assume $h(p,u) \gg 0$ single-valued and differentiable. Write $e(p,u) = p \cdot h(p,u)$ and differentiate in p:

$$\nabla_p e(p,u) \;=\; h(p,u) \;+\; \left[D_p h(p,u)\right]^{\mathsf{T}} p.$$

The second term vanishes:

- EMP first-order conditions: $p = \lambda\, \nabla u\big(h(p,u)\big)$ for some $\lambda \geq 0$.
- "No excess utility" holds identically in p: $u\big(h(p,u)\big) = u$. Differentiate in p: $\left[D_p h\right]^{\mathsf{T}} \nabla u(h) = 0$.
- Hence $\left[D_p h\right]^{\mathsf{T}} p = \lambda \left[D_p h\right]^{\mathsf{T}} \nabla u(h) = 0$, and $\nabla_p e = h$. ■

The envelope logic made exact: re-optimization ($D_p h$) is orthogonal to the price vector, so only the direct effect — the bundle itself — remains.

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Proof: properties of $D_p h$ (MWG 3.G.2)

- *(i)* Shephard: $h(p, u) = \nabla_p e(p, u)$ identically. Differentiate in p: $D_p h = D_p^2 e$.

- *(ii)* $e(\cdot, u)$ is twice continuously differentiable here, so its Hessian is **symmetric** (Young's theorem); and e is **concave** in p (Lecture 2), so its Hessian is **negative semidefinite**. Both properties transfer to $D_p h$ by (i).

- *(iii)* $h(\cdot, u)$ is homogeneous of degree zero in p (Lecture 2). Differentiate $h(\alpha p, u) = h(p, u)$ in α at $\alpha = 1$ (Euler): $D_p h(p, u)\, p = 0$. ■

NSD here is the differential form of the compensated law: $dp \cdot dh = dp^{\mathsf{T}} D_p h\, dp \leq 0$.

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Proof: the Slutsky equation (MWG 3.G.3)

Fix (p, w) and let $u = v(p, w)$; duality gives the identity, in p:

$$h_\ell(p, u) \;=\; x_\ell\big(p,\; e(p, u)\big).$$

Differentiate both sides with respect to p_k (chain rule on the second argument):

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, e(p, u))}{\partial p_k} + \frac{\partial x_\ell(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_k}.$$

Two substitutions finish it: by Shephard, $\partial e/\partial p_k = h_k(p, u)$; and at $u = v(p, w)$ we have $e(p, u) = w$ and $h_k(p, u) = x_k(p, w)$. Hence

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w}\, x_k(p, w) \;=\; s_{\ell k}(p, w). \blacksquare$$

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Proof: Roy's identity (MWG 3.G.4)

Let $\bar{u} = v(\bar{p}, \bar{w})$. Duality gives the identity, in p:

$$v\big(p,\ e(p, \bar{u})\big) \;=\; \bar{u}.$$

Differentiate with respect to p_ℓ at $p = \bar{p}$ (chain rule):

$$\frac{\partial v(\bar{p}, \bar{w})}{\partial p_\ell} + \frac{\partial v(\bar{p}, \bar{w})}{\partial w} \cdot \frac{\partial e(\bar{p}, \bar{u})}{\partial p_\ell} \;=\; 0.$$

By Shephard, $\partial e / \partial p_\ell = h_\ell(\bar{p}, \bar{u}) = x_\ell(\bar{p}, \bar{w})$ (duality at $\bar{u} = v(\bar{p}, \bar{w})$). Since $\partial v / \partial w > 0$:

$$x_\ell(\bar{p}, \bar{w}) \;=\; -\frac{\partial v(\bar{p}, \bar{w}) / \partial p_\ell}{\partial v(\bar{p}, \bar{w}) / \partial w}. \blacksquare$$

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Proof: no wealth effects $\Rightarrow$ CV $=$ ΔCS $=$ EV

Suppose $\partial x_1(p,w)/\partial w = 0$ on the relevant range. Use $h_1(p,u) = x_1\big(p, e(p,u)\big)$ and differentiate in u:

$$\frac{\partial h_1(p,u)}{\partial u} = \frac{\partial x_1}{\partial w} \cdot \frac{\partial e(p,u)}{\partial u} = 0.$$

- So h_1 does not depend on u at all: $h_1(p,u^0) = h_1(p,u^1) = x_1(p,w)$ for every p on the path.
- The three areas are integrals of the *same* function:

$$-CV = \int_{p_1^0}^{p_1^1} h_1(t,u^0)\,dt = \int_{p_1^0}^{p_1^1} x_1(t,w)\,dt = \int_{p_1^0}^{p_1^1} h_1(t,u^1)\,dt = -EV. \blacksquare$$

Quasilinear utility $u = x_1$ ·-free numeraire form delivers this globally; small budget shares deliver it approximately. EC1's quasilinear problem is this theorem in disguise.

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